

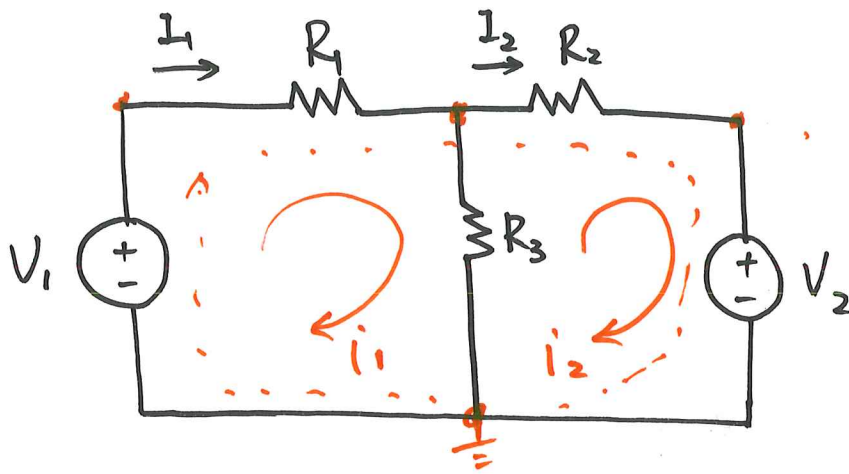
# Mesh Analysis

2017-08-07: P<sub>1</sub>

1. A mesh is a loop that not include any other loops in it.

2. Apply KVL to find unknown currents.

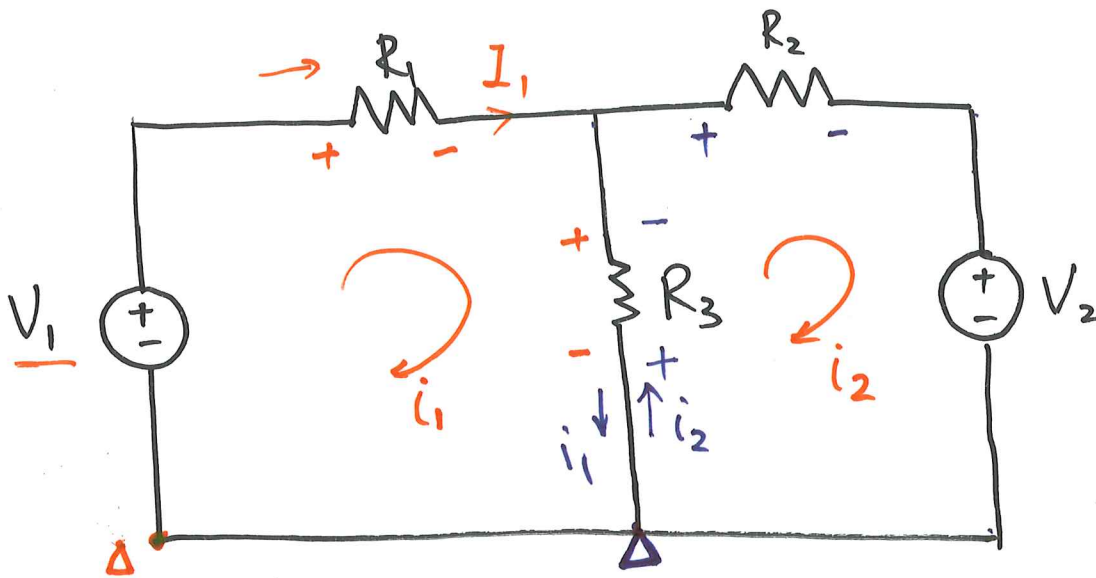
3.



Step 1: assign mesh current

Step 2: KVL  $\rightarrow$  equations

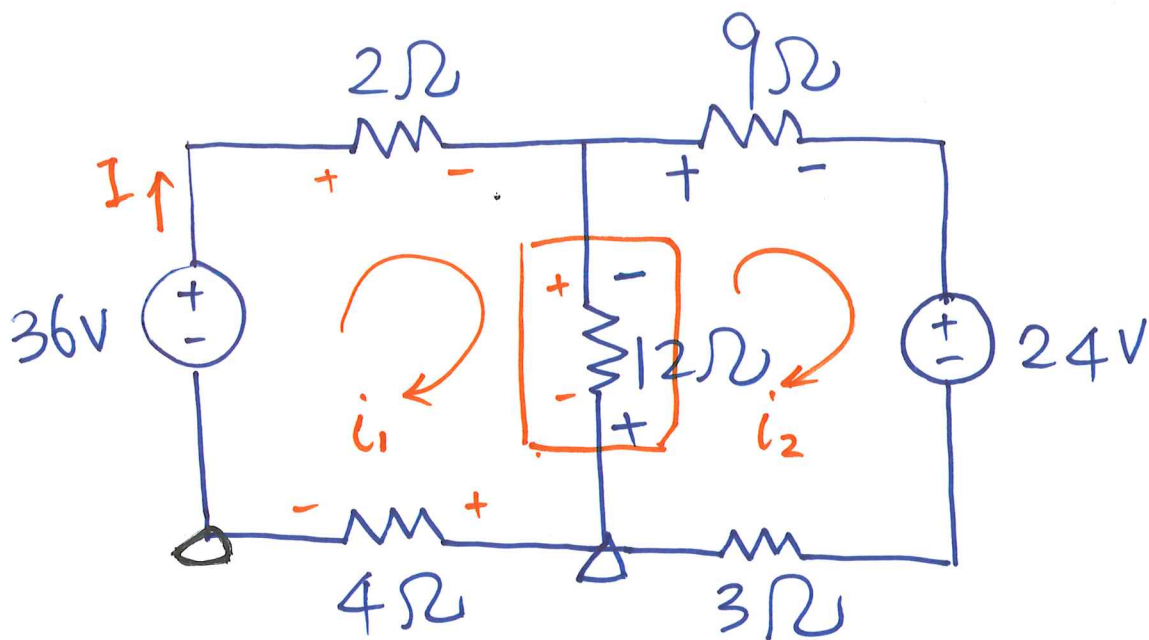
Step 3: Solve for the unknown.



$$\text{Eq 1: } -V_1 + i_1 R_1 + \underbrace{(i_1 - i_2) R_3}_{\text{net current}} = 0$$

$$\text{Eq 2: } (i_2 - i_1) R_3 + i_2 R_2 + V_2 = 0$$

Step 3  $\rightarrow$



$$\text{Mesh 1: } -36 + 2i_1 + (i_1 - i_2) \times 12 + 4i_1 = 0 \quad (1)$$

$$\text{Mesh 2: } 12(i_2 - i_1) + 9i_2 + 24 + 3i_2 = 0 \quad (2)$$

$$(1) \quad 18i_1 - 12i_2 = 36 \Rightarrow \underline{3i_1 - 2i_2 = 6} \rightarrow$$

$$(2) \quad -12i_1 + 24i_2 = -24 \Rightarrow -i_1 + 2i_2 = -2$$

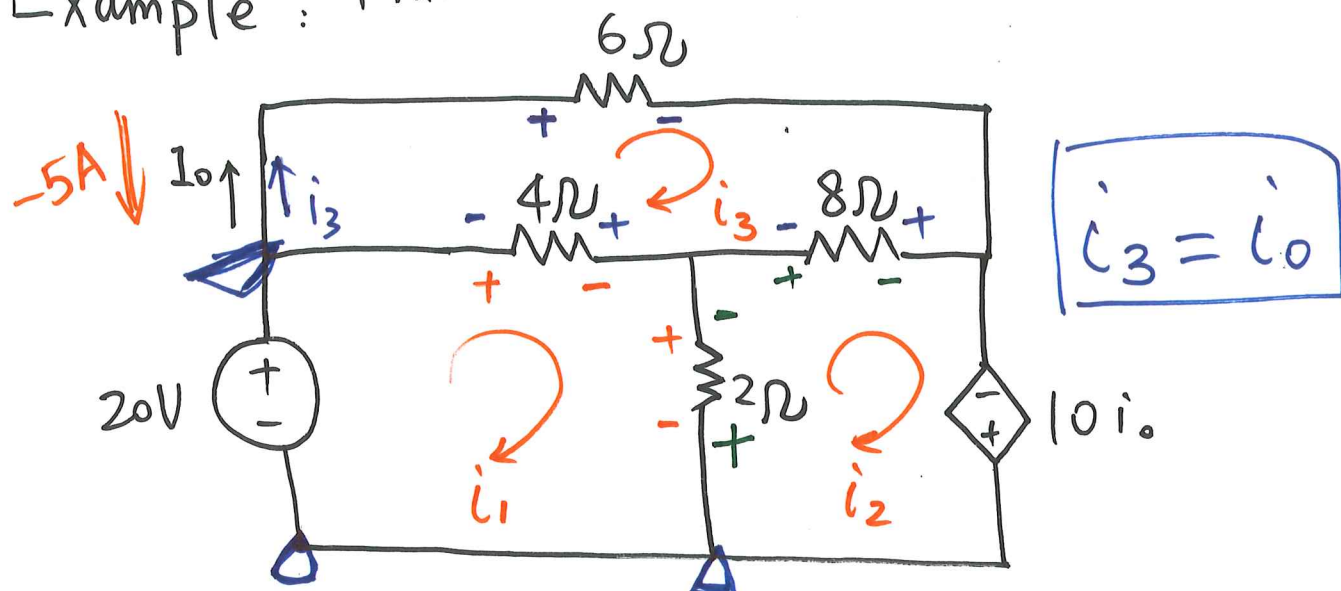
$$i_1 = 2i_2 + 2 \rightarrow$$

$$\underline{6i_2 + 6} - 2i_2 = 6 \Rightarrow$$

$$\boxed{\begin{array}{l} i_2 = 0 \text{ A} \\ i_1 = 2 \text{ A} \end{array}}$$

Example : Find  $I_o$

P<sub>3</sub>



$$\text{Mesh 1: } -20 + 4(i_1 - i_3) + 2(i_1 - i_2) = 0 \quad (1)$$

$$\text{Mesh 2: } 2(i_2 - i_1) + 8(i_2 - i_3) - 10i_3 = 0 \quad (2)$$

$$\text{Mesh 3: } 6i_3 + 8(i_3 - i_2) + 4(i_3 - i_1) = 0 \quad (3)$$

$$(1) \quad 6i_1 - 2i_2 - 4i_3 = 20$$

$$(2) \quad -2i_1 + 10i_2 - 18i_3 = 0$$

$$(3) \quad -4i_1 - 8i_2 + 18i_3 = 0$$

$$\begin{pmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 20 \\ 0 \\ 0 \end{pmatrix}$$

$$i_3 = \frac{\Delta_3}{\Delta} = \frac{\begin{vmatrix} 6 & -2 & 20 \\ -2 & 10 & 0 \\ -4 & -8 & 0 \end{vmatrix} \begin{matrix} \overline{6} & -2 \\ -2 & 10 \\ -4 & -8 \end{matrix}}{\begin{vmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{vmatrix} \begin{matrix} \overline{6} & -2 \\ -2 & 10 \\ -4 & -8 \end{matrix}}$$

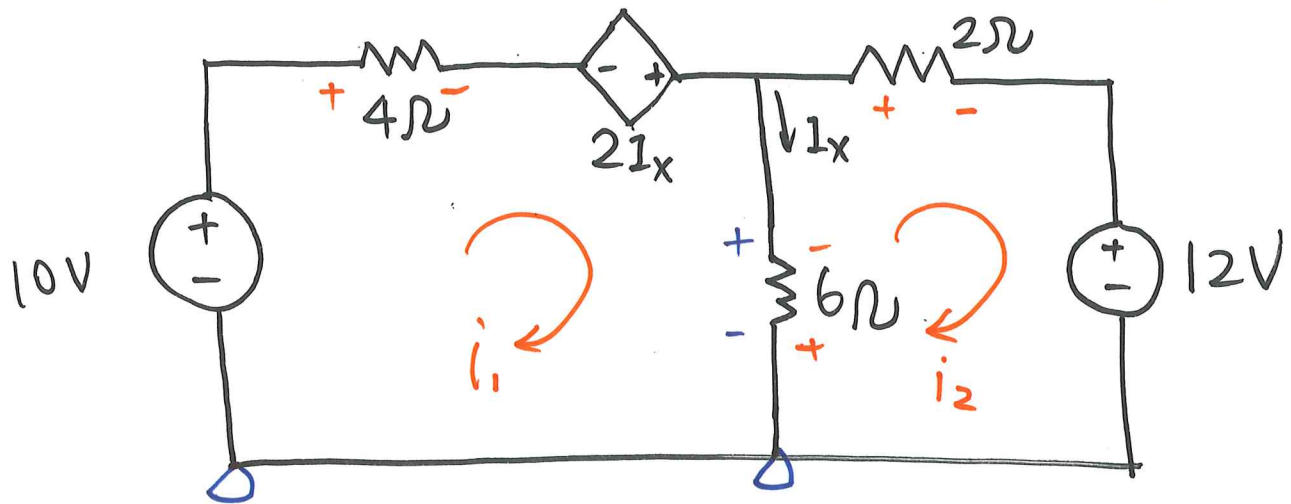
$$= \frac{20 \times 2 \times 8 + 20 \times 10 \times 4}{6 \times 10 \times 18 + 8 \cdot (-18) + 8 \cdot (-8) - 16 \times 10 - 48 \times 18 - 4 \times 18}$$

$$= \frac{1120}{-224} = -5A$$

Example: Determine  $i_1$  and  $i_2$ .

CCVS

P4



$$\text{Mesh 1: } -10 + 4i_1 - \underline{2I_x} + 6(i_1 - i_2) = 0 \quad (1)$$

$$\text{Mesh 2: } 6(i_2 - i_1) + 2i_2 + 12 = 0 \quad (2)$$

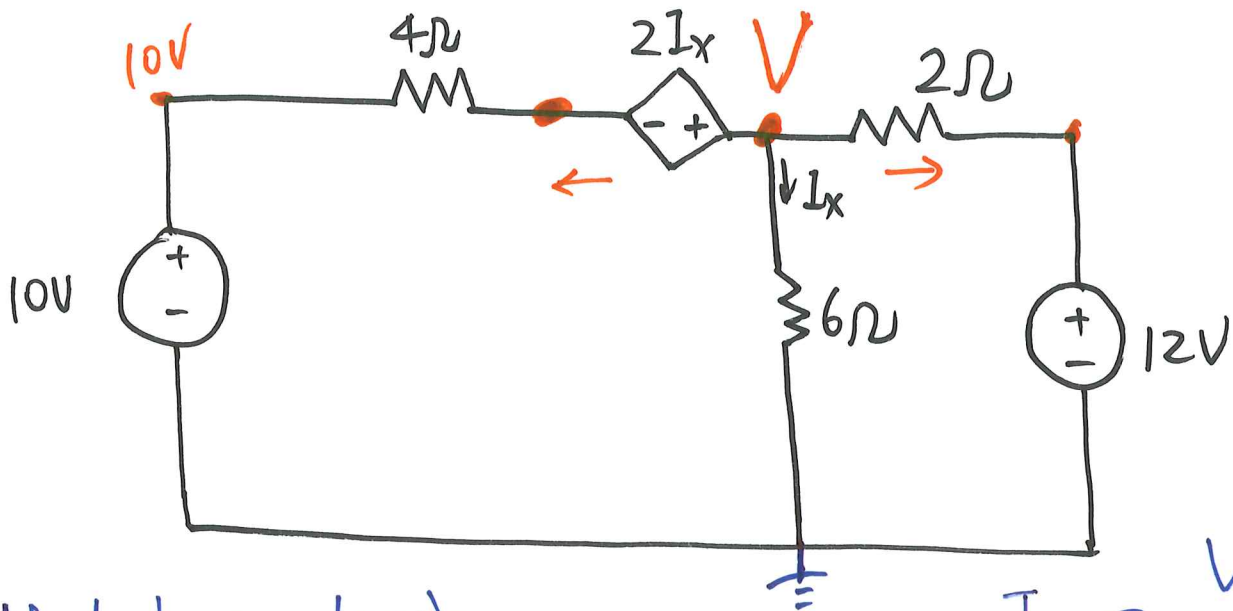
$$I_x: I_x = i_1 - i_2$$

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$$i_1 = 0.8 \text{ A}, \quad i_2 = -0.9 \text{ A}$$

$$I_x = 1.7 \text{ A}$$





Nodal analysis's:

$$I_x = \frac{V}{6} \quad (2)$$

$$\frac{V - 2I_x - 10}{4} + \frac{V - 12}{2} + \frac{V}{6} = 0 \quad (1)$$

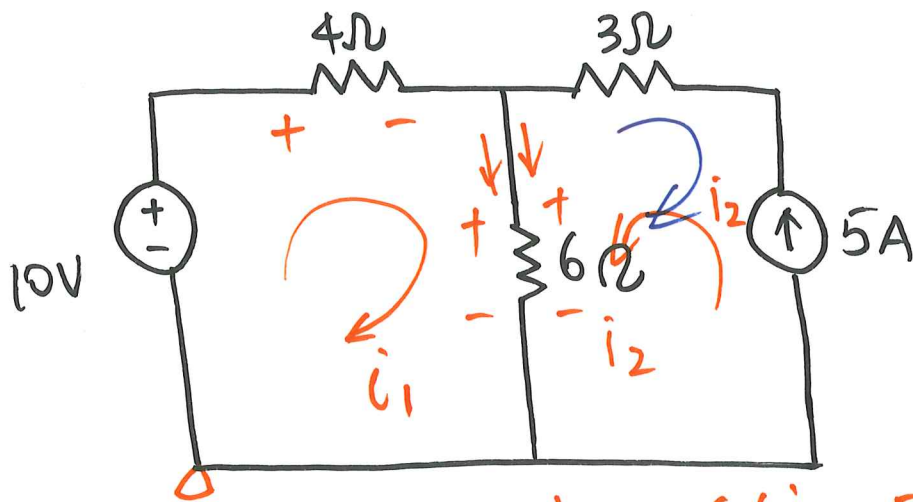
$$\textcircled{1} \times 12: 3V - 2 \cdot \frac{V}{6} - 30 + 6V - 72 + 2V = 0$$

$$10V = 102 \Rightarrow V = 10.2V.$$

$$I_x = \frac{10.2}{6} A = 1.7 A$$

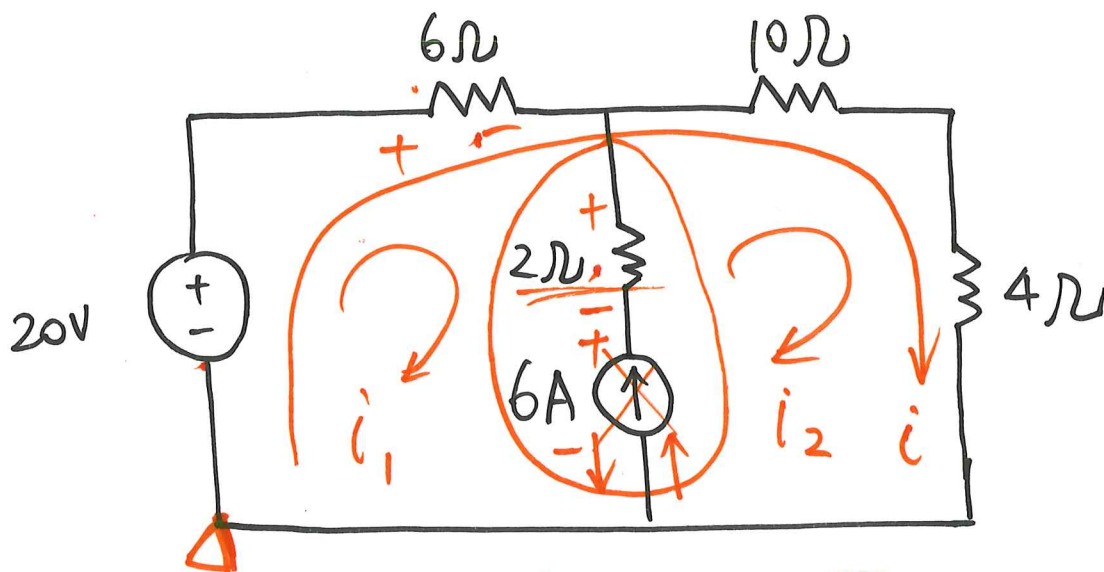
# Supermesh

P<sub>1</sub>



$$\underline{i_2 = 5A}$$

Mesh 1:  $-10 + 4i_1 + 6(i_1 + 5) = 0$   
 $10i_1 = -20 \Rightarrow i_1 = -2A$



$$6 = i_2 - i_1 \quad (1)$$

Mesh:  $-20 + 6i_1 + 2(i_1 - i_2) + \dots = 0 \text{ KVL}$

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0 \quad (2)$$



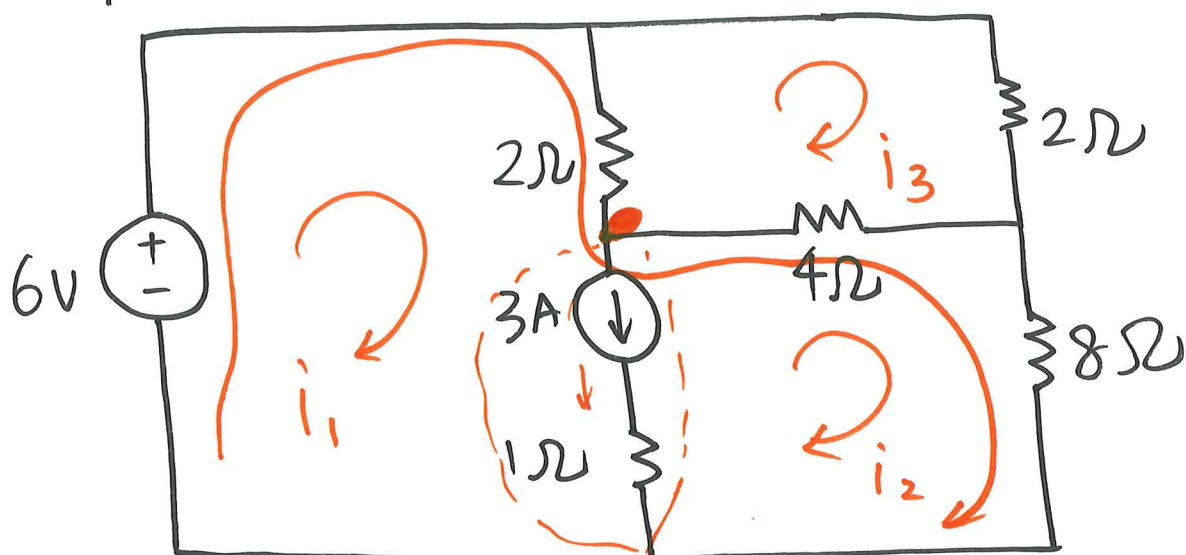
$$-10 + 4i_1 + 6(i_1 - i_2) = \cancel{30} 0$$

$$-10 + 10i_1 - \underset{(-5)}{6i_2} = 0, \quad i_2 = -5A$$

$$10i_1 = -20 \Rightarrow i_1 = -2A$$

Example: Determine mesh current.

P<sub>2</sub>



Branch current:  $3 = i_1 - i_2$ ; (1)

Supermesh:

$$-6 + 2(i_1 - i_3) + 4(i_2 - i_3) + 8i_2 = 0 \quad (2)$$

$$\text{Mesh 3: } 2(i_3 - i_1) + 2i_3 + 4(i_3 - i_2) = 0 \quad (3)$$

$$(1) \quad i_1 - i_2 = 3$$

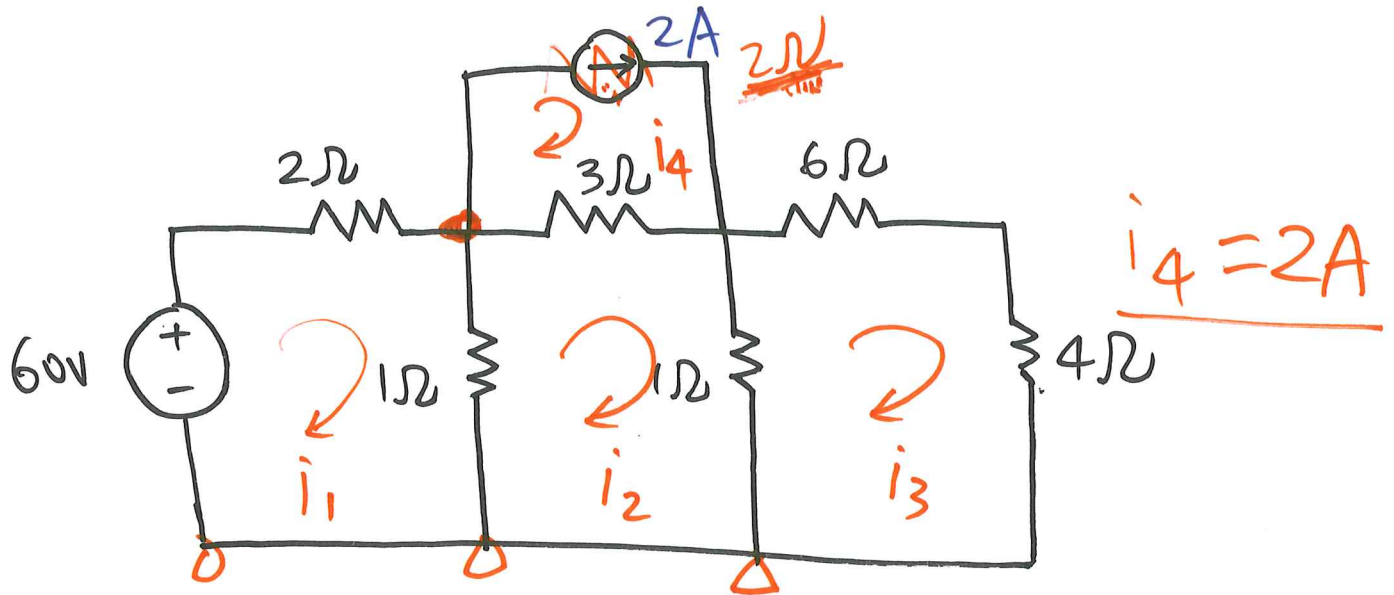
$$(2) \quad 2i_1 + 12i_2 - 6i_3 = 6$$

$$(3) \quad -2i_1 - 4i_2 + 8i_3 = 0$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 2 & 12 & -6 \\ -2 & -4 & 8 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

Example: Determine mesh current.

P3



$$\textcircled{1} -60 + 2i_1 + (i_1 - i_2) = 0$$

$$\textcircled{2} (i_2 - i_1) + 3(i_2 - 2) + (i_2 - i_3) = 0$$

$$\textcircled{3} (i_3 - i_2) + 6i_3 + 4i_3 = 0$$

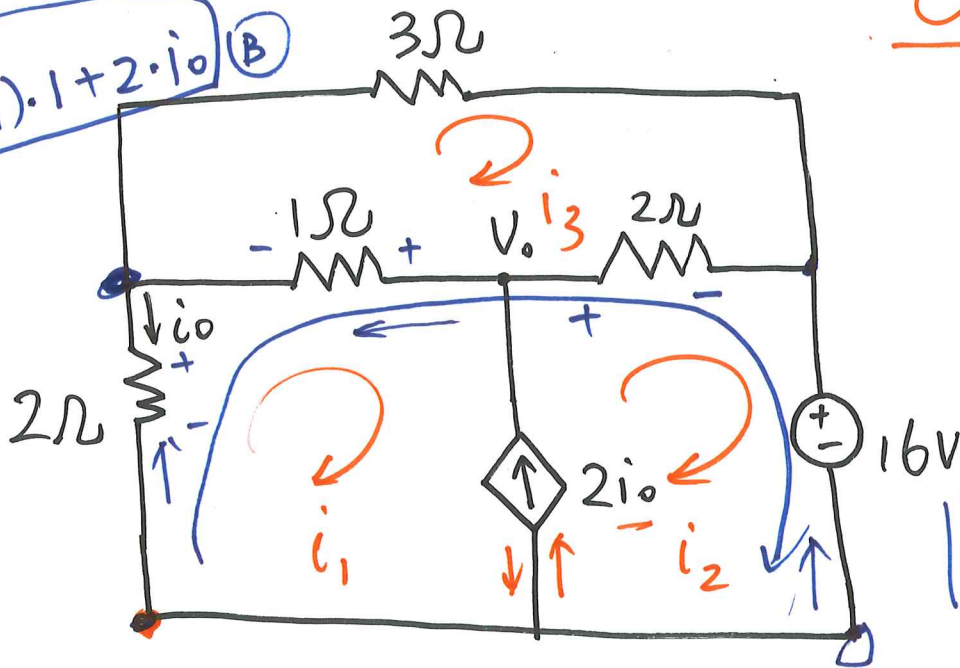
$$\textcircled{4} 2i_4 + 3(i_4 - i_2) = 0$$

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Example: Find  $V_o$  and  $i_o$ .

P4

CCCS



$$i_o = -i_1$$

$$\textcircled{A} \quad V_o = 16 + 2(i_2 - i_3)$$

Branch current:  $2i_o = i_2 - i_1$   
 $-2i_1$

$\textcircled{1}$

Supermesh:

$$2i_1 + 1(i_1 - i_3) + 2(i_2 - i_3) + 16 = 0 \quad \textcircled{2}$$

$$\text{Mesh 3: } 3i_3 + 2(i_3 - i_2) + 1(i_3 - i_1) = 0 \quad \textcircled{3}$$

$$i_1 + i_2 = 0 \quad \textcircled{1}$$

$$3i_1 + 2i_2 + (-3)i_3 = -16$$

$$i_1 + 2(\cancel{i_1} + i_2) - 3i_3 = -16 \Rightarrow i_1 - 3i_3 = -16 \quad \textcircled{2}$$

$$\cancel{-i_1} - \cancel{i_2} - i_2 + 6i_3 = 0 \Rightarrow i_2 = 6i_3 \quad \textcircled{3}$$

$$\begin{array}{l} i_1 = 3i_3 - 16 \\ i_2 = 6i_3 \end{array} \quad \left| \quad 9i_3 = 16 \Rightarrow i_3 = \frac{16}{9} \right. \nearrow$$